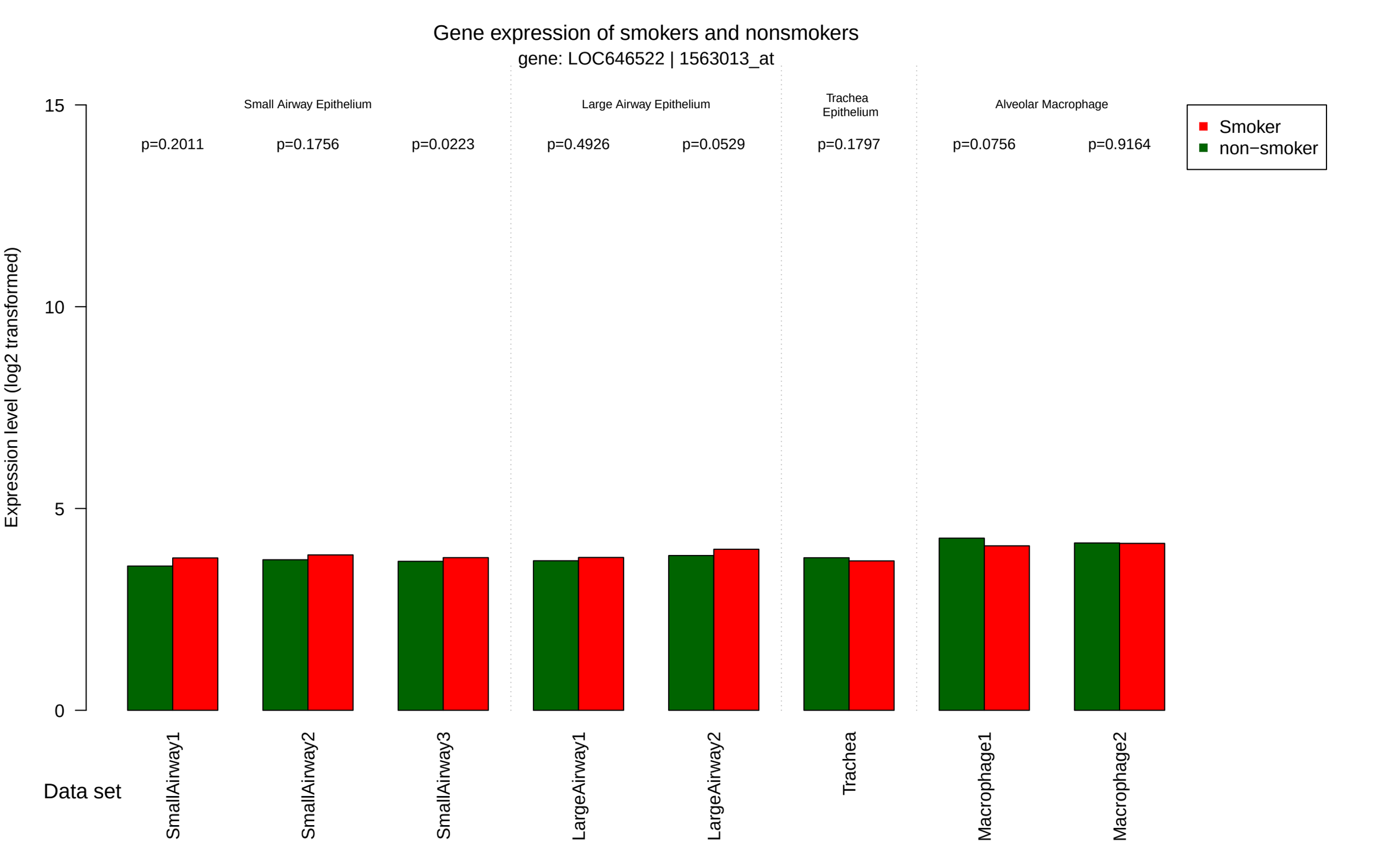
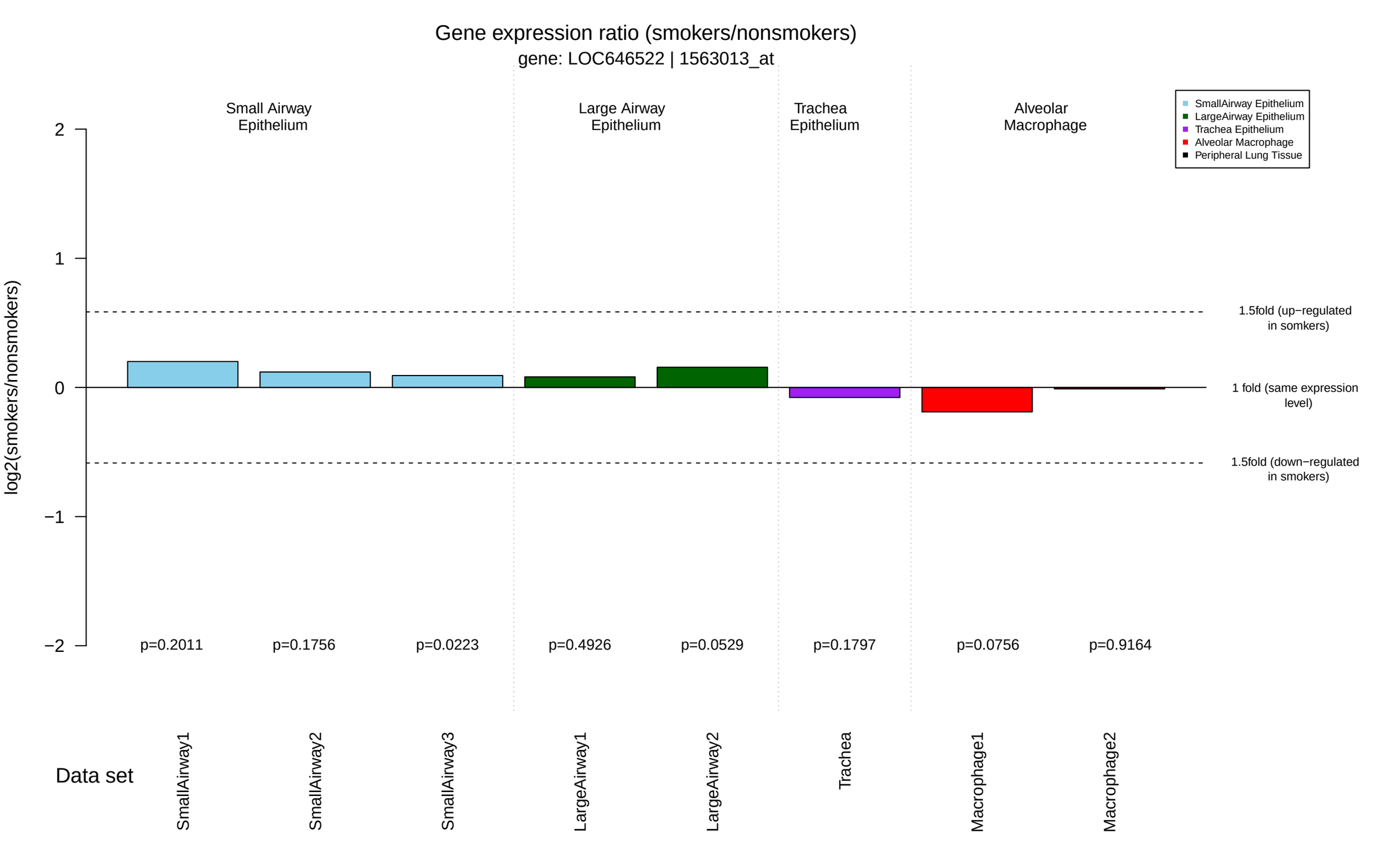
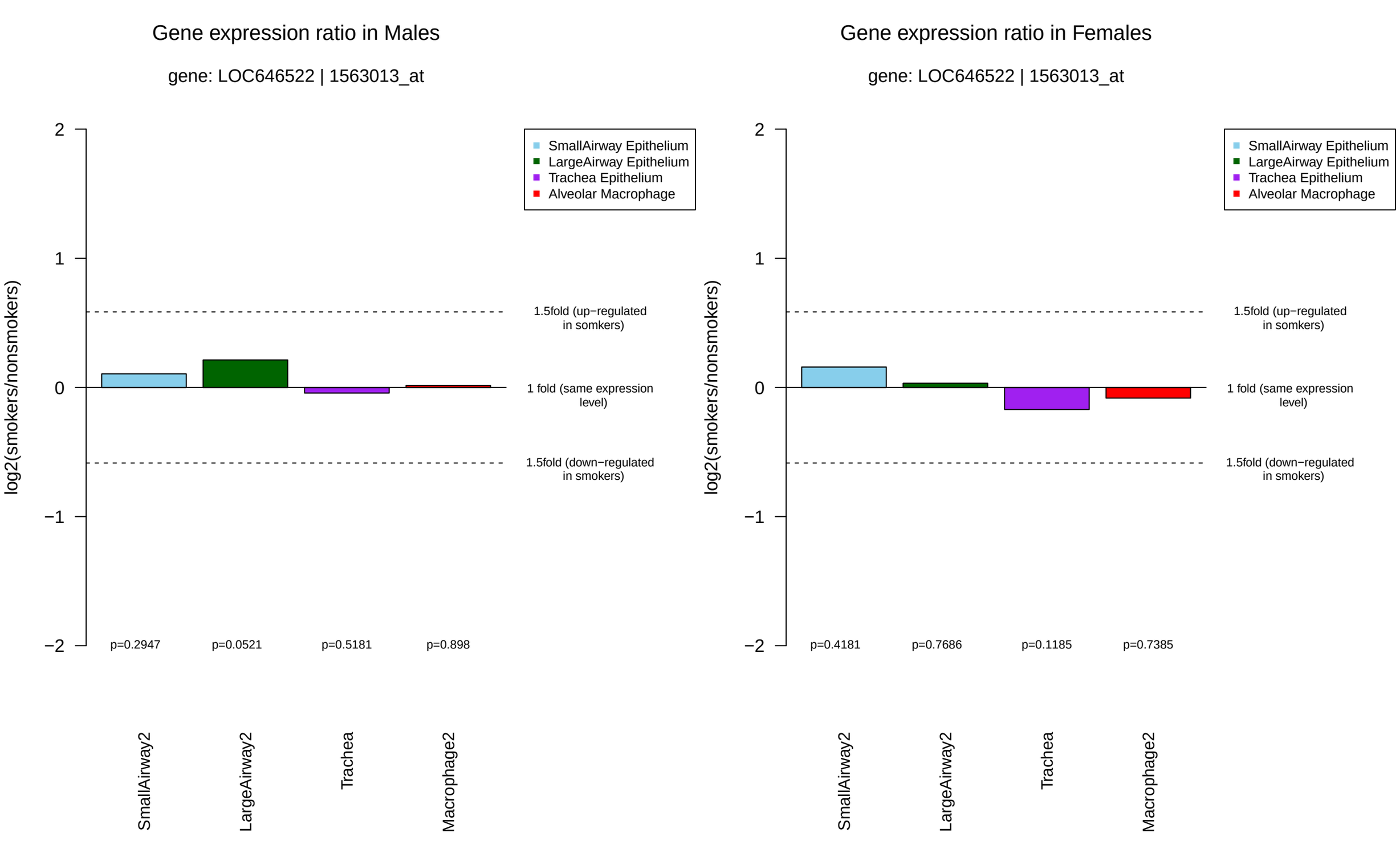
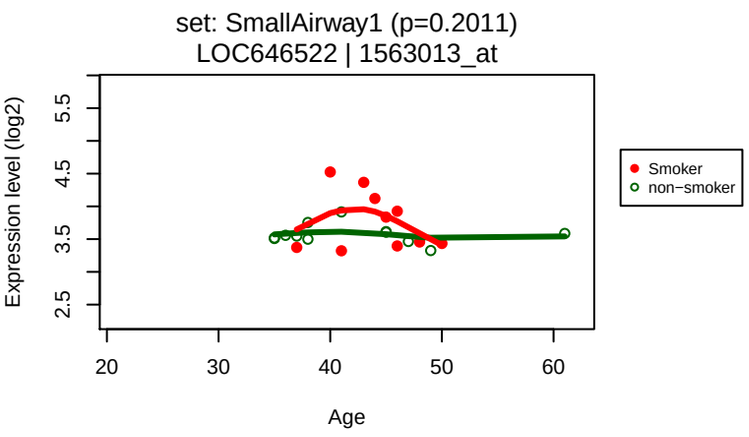
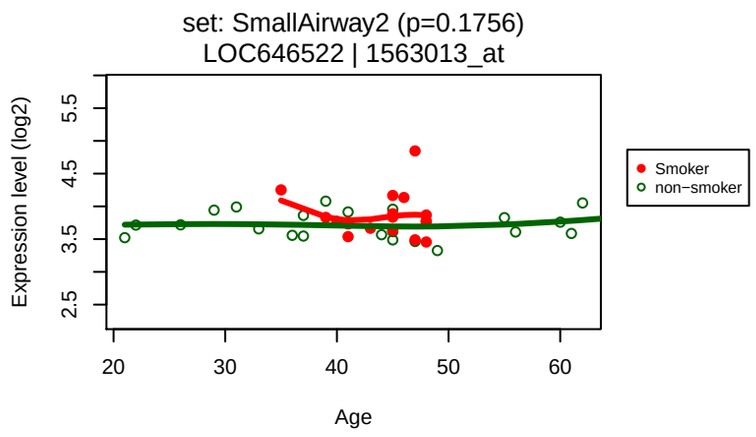
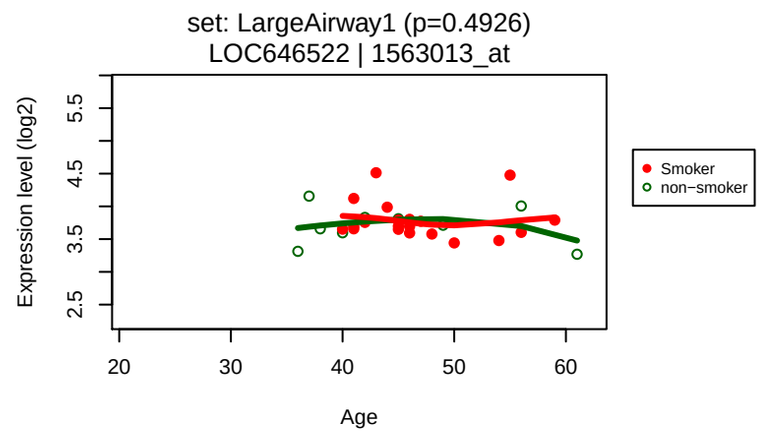

Gene expression of smokers and nonsmokers
gene: LOC646522 | 1563013_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2011
p=0.1756
p=0.0223
p=0.4926
p=0.0529
p=0.1797
p=0.0756
p=0.9164
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

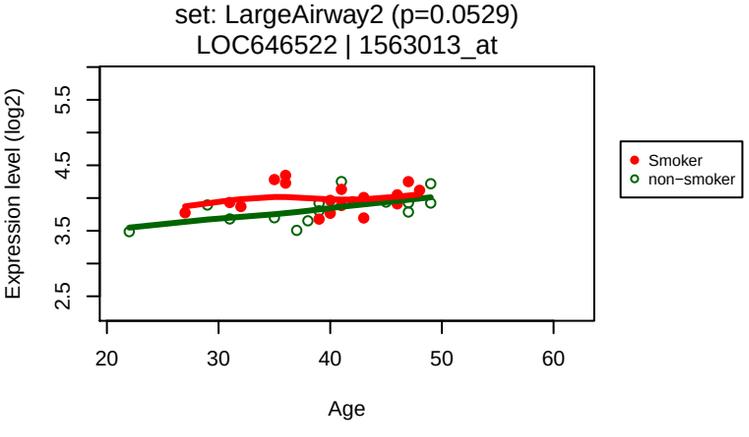

Gene expression ratio (smokers/nonsmokers)
gene: LOC646522 | 1563013_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2011
p=0.1756
p=0.0223
p=0.4926
p=0.0529
p=0.1797
p=0.0756
p=0.9164
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

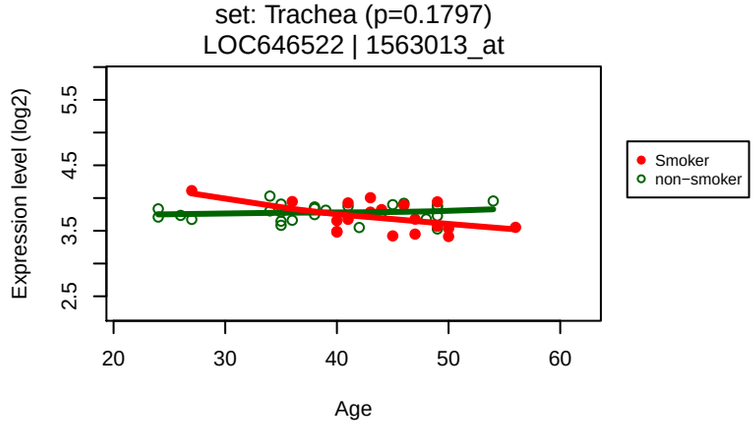

Gene expression ratio in Males
gene: LOC646522 | 1563013_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2947
p=0.0521
p=0.5181
p=0.898
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC646522 | 1563013_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4181
p=0.7686
p=0.1185
p=0.7385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

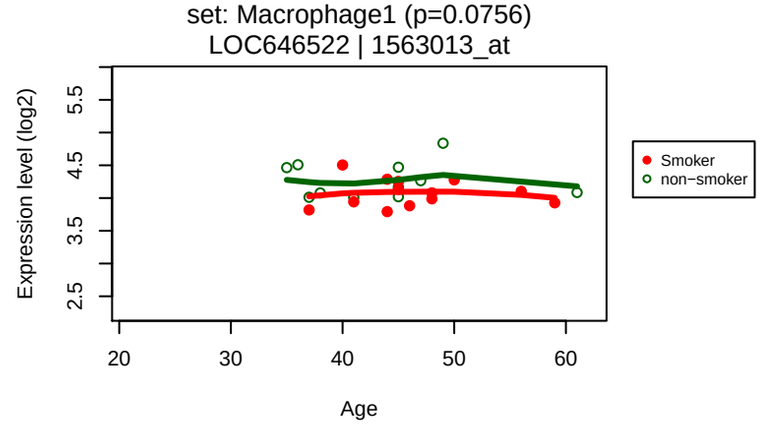

set: SmallAirway1 (p=0.2011)
LOC646522 | 1563013_at
Expression level (log2)
Age
Smoker
non−smoker

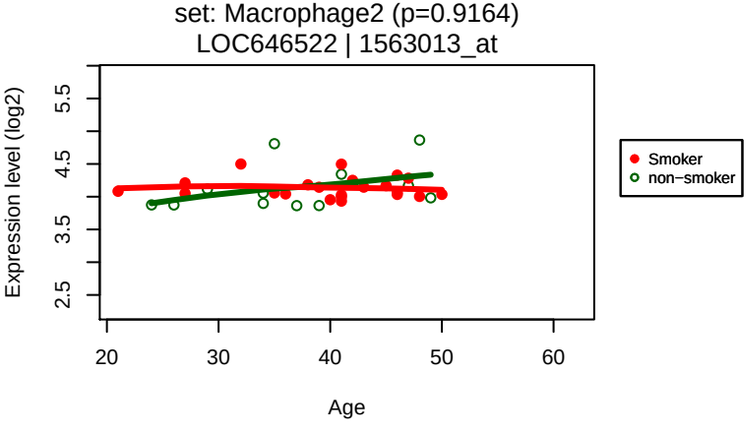

set: SmallAirway2 (p=0.1756)
LOC646522 | 1563013_at
Expression level (log2)
Age
Smoker
non−smoker

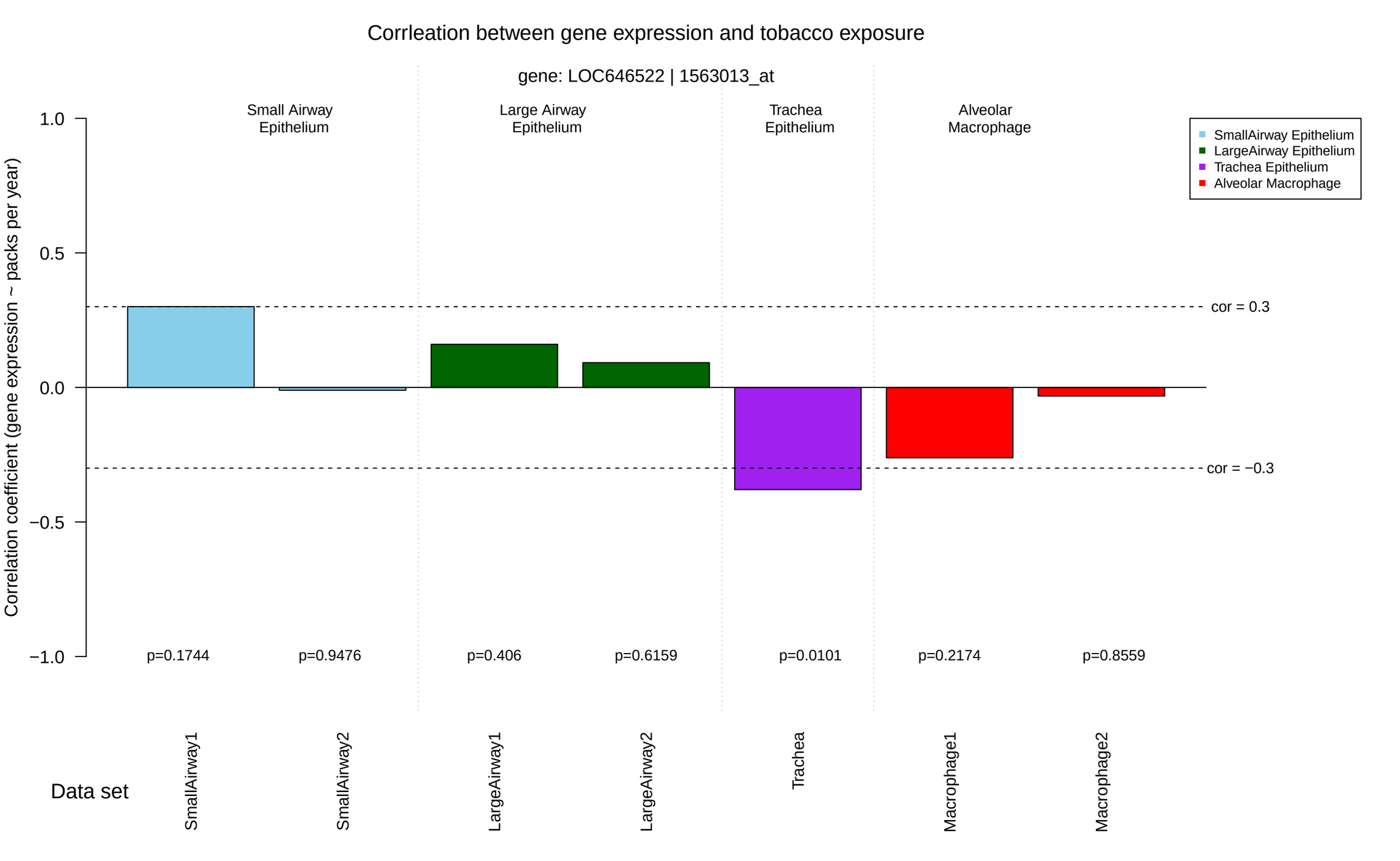

set: LargeAirway1 (p=0.4926)
LOC646522 | 1563013_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0529)
LOC646522 | 1563013_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1797)
LOC646522 | 1563013_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0756)
LOC646522 | 1563013_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9164)
LOC646522 | 1563013_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC646522 | 1563013_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1744
p=0.9476
p=0.406
p=0.6159
p=0.0101
p=0.2174
p=0.8559
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2